

Matrix Factorization via Poisson SuSiE

1 Model

We observe a count matrix $Y \in \mathbb{N}^{N \times M}$. Each row y_i is modeled as

$$y_{im} \sim \text{Poi}\left(\sum_{d=1}^D \beta_{id} F_{\gamma_{id}, m}\right), \quad m \in [M],$$

where $F \in \mathbb{R}_+^{K \times M}$ is a shared factor matrix satisfying $\sum_m F_{km} = 1$ for all k , and for each observation i and effect d :

$$\gamma_{id} \sim \text{Multi}(1, \frac{1}{K} \mathbf{1}_K), \tag{1}$$

$$\beta_{id} \mid \gamma_{id} \sim \text{Gam}(\alpha_{id}^0, \lambda_{id}^0). \tag{2}$$

2 Variational family

We use the mean-field variational family

$$q(\beta, \gamma) = \prod_{i=1}^N \prod_{d=1}^D q(\gamma_{id}) q(\beta_{id} \mid \gamma_{id}),$$

where $q(\gamma_{id}) = \text{Multi}(1, \bar{\gamma}_{id})$ with $\bar{\gamma}_{id} = (\bar{\gamma}_{id1}, \dots, \bar{\gamma}_{idK})$, and $q(\beta_{id} \mid \gamma_{id} = k) = \text{Gam}(\alpha_{idk}, \lambda_{idk})$.

We also introduce auxiliary variables $\xi_{imd} \geq 0$ with $\sum_d \xi_{imd} = 1$ for each (i, m) .

3 Variational parameters

The optimal variational parameters are:

$$\alpha_{idk} = \sum_m \xi_{imd} y_{im} + \alpha_{id}^0, \tag{3}$$

$$\lambda_{idk} = \sum_m F_{km} + \lambda_{id}^0. \tag{4}$$

Note that α_{idk} does not depend on k , and λ_{idk} does not depend on i .

3.1 Moments

Under $q(\beta_{id} \mid \gamma_{id} = k) = \text{Gam}(\alpha_{idk}, \lambda_{idk})$:

$$\mathbb{E}_{q(\beta_{id} \mid \gamma_{id} = k)}[\log \beta_{id}] = \psi(\alpha_{idk}) - \log(\lambda_{idk}), \tag{5}$$

$$\mathbb{E}_{q(\beta_{id}, \gamma_{id})}[\log \beta_{id}] = \sum_k \bar{\gamma}_{idk} [\psi(\alpha_{idk}) - \log(\lambda_{idk})]. \tag{6}$$

4 ELBO

The evidence lower bound is

$$\begin{aligned} \mathcal{E}(q) = & \sum_{i=1}^N \sum_{d=1}^D \mathbb{E}_{q_{id}(\gamma_{id}, \beta_{id})} \left[\sum_m (\xi_{imd} y_{im} (\log \beta_{id} + \log F_{\gamma_{id}, m}) - \beta_{id} F_{\gamma_{id}, m}) \right] \\ & - \sum_{i=1}^N \sum_{d=1}^D \text{KL}(q(\gamma_{id}, \beta_{id}) \| p(\gamma_{id}, \beta_{id})) - \sum_{i,m,d} y_{im} \xi_{imd} \log \xi_{imd}. \end{aligned} \quad (7)$$

Expanding term by term:

$$\text{Term 1} = \sum_{i,d,m} \xi_{imd} y_{im} \mathbb{E}_q[\log \beta_{id}], \quad (8)$$

$$\text{Term 2} = \sum_{i,d,m} \xi_{imd} y_{im} \sum_k \bar{\gamma}_{idk} \log F_{km}, \quad (9)$$

$$\text{Term 3} = - \sum_{i,d,m} \sum_k \bar{\gamma}_{idk} \frac{\alpha_{idk}}{\lambda_{idk}} F_{km}, \quad (10)$$

$$\text{Term 4} = - \sum_{i,m,d} y_{im} \xi_{imd} \log \xi_{imd}. \quad (11)$$

The KL divergence decomposes as $\text{KL} = \text{KL}_{\text{disc}} + \text{KL}_{\text{cont}}$:

$$\text{KL}_{\text{disc}} = \sum_{i,d,k} \bar{\gamma}_{idk} (\log \bar{\gamma}_{idk} - \log \frac{1}{K}), \quad (12)$$

$$\begin{aligned} \text{KL}_{\text{cont}} = & \sum_{i,d,k} \bar{\gamma}_{idk} \left[(\alpha_{idk} - \alpha_{id}^0) \psi(\alpha_{idk}) + (\log \lambda_{idk} - \log \lambda_{id}^0) \alpha_{id}^0 \right. \\ & \left. - \log \frac{\Gamma(\alpha_{idk})}{\Gamma(\alpha_{id}^0)} - (\lambda_{idk} - \lambda_{id}^0) \frac{\alpha_{idk}}{\lambda_{idk}} \right]. \end{aligned} \quad (13)$$

5 Algorithm

Algorithm 1 MF-Poisson-SuSiE

- 1: **for** iter = 1, ..., T **do**
- 2: **(Break symmetry)** Merge duplicate effects if needed.
- 3: **for** $d = 1, \dots, D$ **do**
- 4: **E-step:** Update ξ_{imd} for all i, m :

$$\xi_{imd} \leftarrow \text{Norm-Exp}\left(\mathbb{E}_{q(\gamma_{id}, \beta_{id})}[\log \beta_{id} + \log F_{\gamma_{id}, m}]\right), \quad \forall i \in [N], m \in [M].$$

- 5: **M-step:** Update variational parameters for effect d :

$$\begin{aligned} \alpha_{idk} &\leftarrow \sum_m \xi_{imd} y_{im} + \alpha_{id}^0, \\ \lambda_{idk} &\leftarrow \sum_m F_{km} + \lambda_{id}^0. \end{aligned}$$

- 6: **PIP:** Compute logprob_{idk} and update $\bar{\gamma}_{id}$:

$$\begin{aligned} \text{logprob}_{idk} &= \sum_m \xi_{imd} y_{im} \log F_{km} - \alpha_{idk} \log \lambda_{idk}, \\ \bar{\gamma}_{id} &\leftarrow \text{Norm-Exp}(\text{logprob}_{id1}, \dots, \text{logprob}_{idK}). \end{aligned}$$

- 7: **VEB** (optional): Update prior hyperparameters $(\alpha_{id}^0, \lambda_{id}^0)$:

$$\begin{aligned} \bar{\beta}_{id} &= \sum_k \bar{\gamma}_{idk} \frac{\alpha_{idk}}{\lambda_{idk}}, \quad \overline{\log \beta}_{id} = \sum_k \bar{\gamma}_{idk} [\psi(\alpha_{idk}) - \log \lambda_{idk}], \\ \lambda_{id}^0 &\leftarrow \frac{\alpha_{id}^0}{\bar{\beta}_{id}}, \\ \alpha_{id}^0 &\leftarrow \psi^{-1}(\overline{\log \beta}_{id} + \log \lambda_{id}^0), \\ \lambda_{id}^0 &\leftarrow \frac{\alpha_{id}^0}{\bar{\beta}_{id}}. \end{aligned}$$

- 8: **end for**
- 9: **Update** F (optional): For each k, m :

$$\begin{aligned} A_{km} &= \sum_{i,d} \bar{\gamma}_{idk} \xi_{imd} y_{im}, \\ F_{km} &\leftarrow \frac{A_{km}}{\sum_{m'} A_{km'}}. \end{aligned}$$

- 10: **end for**
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6 Derivation of the F update

The terms in the ELBO involving F_{km} are:

$$\sum_{k,m} \left[A_{km} \log F_{km} - B_k F_{km} \right],$$

where

$$A_{km} = \sum_{i,d} \bar{\gamma}_{idk} \xi_{imd} y_{im}, \quad B_k = \sum_{i,d} \bar{\gamma}_{idk} \frac{\alpha_{idk}}{\lambda_{idk}}.$$

Maximizing subject to $\sum_m F_{km} = 1$ via Lagrange multipliers:

$$\frac{\partial}{\partial F_{km}} \left[A_{km} \log F_{km} - B_k F_{km} - \mu_k \left(\sum_m F_{km} - 1 \right) \right] = 0 \implies F_{km} = \frac{A_{km}}{B_k + \mu_k}.$$

Applying the constraint: $B_k + \mu_k = \sum_m A_{km}$, hence

$$F_{km} = \frac{A_{km}}{\sum_{m'} A_{km'}}.$$

7 Derivation of the VEB updates

The expected log-prior for effect d of observation i is:

$$\mathbb{E}_q[\log p(\beta_{id} \mid \gamma_{id}; \alpha_{id}^0, \lambda_{id}^0)] = \sum_k \bar{\gamma}_{idk} \left[\alpha_{id}^0 \log \lambda_{id}^0 - \log \Gamma(\alpha_{id}^0) + (\alpha_{id}^0 - 1)(\psi(\alpha_{idk}) - \log \lambda_{idk}) - \lambda_{id}^0 \frac{\alpha_{idk}}{\lambda_{idk}} \right].$$

λ_{id}^0 **update (closed form)**. Setting $\frac{\partial}{\partial \lambda_{id}^0} = 0$:

$$\sum_k \bar{\gamma}_{idk} \left[\frac{\alpha_{id}^0}{\lambda_{id}^0} - \frac{\alpha_{idk}}{\lambda_{idk}} \right] = 0 \implies \lambda_{id}^0 = \frac{\alpha_{id}^0}{\sum_k \bar{\gamma}_{idk} \alpha_{idk} / \lambda_{idk}} = \frac{\alpha_{id}^0}{\bar{\beta}_{id}}.$$

α_{id}^0 **update (implicit equation)**. Setting $\frac{\partial}{\partial \alpha_{id}^0} = 0$:

$$\sum_k \bar{\gamma}_{idk} \left[\log \lambda_{id}^0 - \psi(\alpha_{id}^0) + \psi(\alpha_{idk}) - \log \lambda_{idk} \right] = 0 \implies \psi(\alpha_{id}^0) = \overline{\log \beta}_{id} + \log \lambda_{id}^0.$$

This is solved by inverting the digamma function, e.g., via Newton's method:

$$\alpha_{id}^0 \leftarrow \alpha_{id}^0 - \frac{\psi(\alpha_{id}^0) - \text{target}}{\psi'(\alpha_{id}^0)}, \quad \text{target} = \overline{\log \beta}_{id} + \log \lambda_{id}^0.$$